

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 22)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

01 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

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01 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1

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2

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01 . 3

Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores

Gas analysis of ice cores

Radiocarbon dating

Magnesium : calcium ratio in calcite

10

0 2 . 1

Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 2 . 2

Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

0 2 . 3

Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

0 2 . 4

Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

10

0 3 . 1

Describe a polyculture system using several species of carp and explain how it increases total productivity.

[4 marks]

0 3 . 2

Explain how integrated multi-trophic aquaculture (IMTA) and aquaponics reduce pollution while increasing food production.

[4 marks]

0 3 . 3

Explain why tuna aquaculture cannot yet replace tuna fishing.

[2 marks]**10**

0 4

Table 1 shows life-history features of three fish species and the age at which they reach a catchable size.

Table 1

Species	Age at catchable size / years	Age at first breeding / years	Eggs per female per year	Maximum lifespan / years
Atlantic cod	2	4	Millions	25
Orange roughy	20	30	Thousands	150
Herring	2	2	Tens of thousands	20

0 4 . 1

Use Table 1 to explain why orange roughy populations collapsed within ten years of large-scale fishing beginning in the late 1970s.

[4 marks]

0 4 . 2

Explain why herring populations recover from overfishing much faster than cod.

[2 marks]

0 4 . 3

Explain why the total mass of fish landed can stay the same while a population is being overfished.

[2 marks]

0 4 . 4

Describe how the Greenland shark and blue (common) skate illustrate the threat of by-catch to k-selected species.

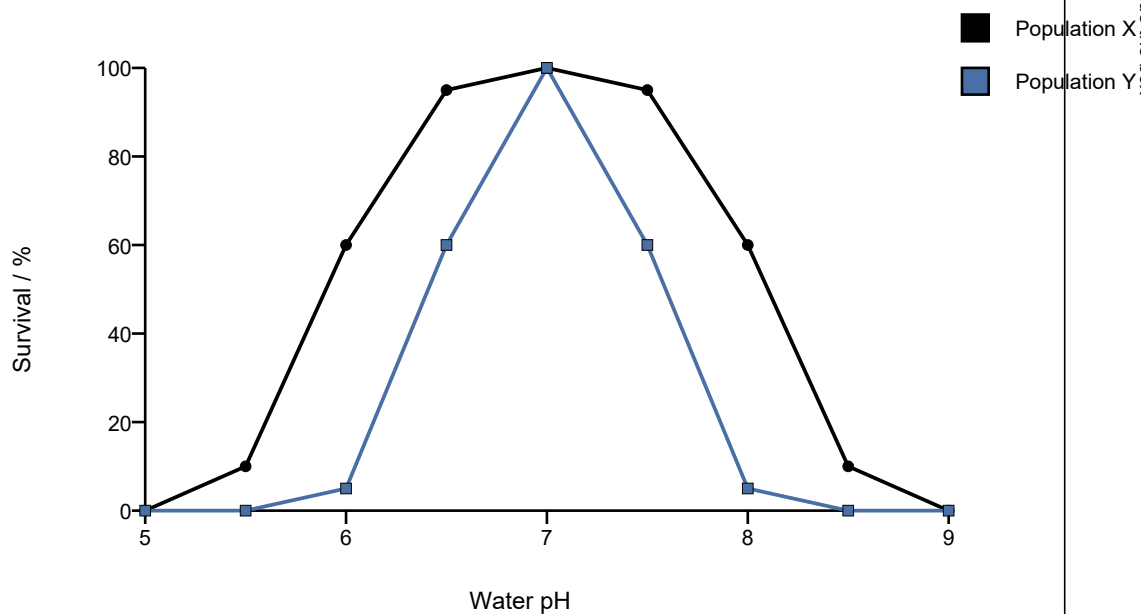
[2 marks]

10

0 5

Figure 1 shows the percentage survival of two populations of a freshwater snail species (X and Y) at different water pH values.

Figure 1



0 5

. 1

Explain what is meant by the term **range of tolerance** and use Figure 1 to compare the ranges of tolerance of populations X and Y.

[3 marks]

0 5

. 2

A stream is acidified by mine drainage from pH 7.0 to 6.0. Use Figure 1 to explain which population is more likely to survive and why.

[3 marks]

0 5 . 3

Explain how an understanding of a species' range of tolerance helps in conservation management.

[2 marks]

0 5 . 4

Give **two** biotic factors that affect the distribution of a species.

[2 marks]

1

2

10

0 6 . 1

A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

- A The readings are precise but may not be accurate
- B The readings are accurate but not precise
- C The readings are both accurate and precise
- D The readings are reliable because the meter is electronic

☐☐☐☐

0 6 . 2

Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

06 . 3

Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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5

07 . 1

Describe the resources obtained from forests. Give **one** example for each category.

[5 marks]

Timber

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Fuel

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Food

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Fibres

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Medicines

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07 . 2

Explain why forests are an important carbon reservoir and how they can be used to counteract climate change.

[3 marks]

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07 . 3

Explain how forests regulate the hydrological cycle and local climate.

[2 marks]

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08 . 1

Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]**Table 1**

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

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08 . 2

Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

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08 . 3

Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.**[2 marks]**

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08 . 4

Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

09

Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

09 . 1

Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

09 . 2

Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

09 . 3

Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

09 . 4

Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

1 0 . 1

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

Do not write outside the box

Extra space

10 . 2 State what is meant by **rewilding**.

[1 mark]

10

Write an essay on one of the following topics.

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Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

[25 marks]

OR

1 1 . 2

Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS